The title should simply and concisely convey the main findings. Avoid nonstandard abbreviations and acronyms.

Ann Author* and Second Author†
Authors' affiliations
Include all institutions where the work was conducted: department or division, institution, city, state (if relevant), and country, in this order.
(CLEO Collaboration)
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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. THEORETICAL FRAMEWORK

In this section, the energy density funcitonal (EDF) employed is outlined.

A. Skyrme EDF

The total energy of the system can be expressed using an EDF by accounting for five different contributions:

$$E = E_{\rm Kin} + E_{\rm Sk} + E_{\rm cm} + E_{\rm Coul} + E_{\rm Pair} \tag{1}$$

$$= \int d\boldsymbol{r}[\mathcal{E}_{\rm Kin} + \mathcal{E}_{\rm Sk} + \mathcal{E}_{\rm cm} + \mathcal{E}_{\rm Coul} + \mathcal{E}_{\rm Pair}], \tag{2}$$

where $E_{\rm Kin}$ is the kinetic energy, $E_{\rm Sk}$ the Skyrme energy, which models the strong interaction between the nucleons, $E_{\rm cm}$ the center or mass correction, $E_{\rm Coul}$ the electric repulsion due to the protons and $E_{\rm Pair}$ the interaction describing pairing correlations.

1. *Skyrme energy functional*

The Skyrme part of the EDF models the strong interaction between the nucleons. From this point forward we will restrict ourselves to the case of time-reversal symmetry of the nucleus. In such case, time-odd densities vanish, and the Skyrme energy reduces to [1–3]

$$\mathcal{E}_{\rm Sk} = \sum_{t=0,1} \Big\{ \; C_t^{\rho}[\rho_0]\rho_t^2 + C_t^{\rho_t}\tau_t + C_t^{\Delta\rho}\rho_t\nabla^2\rho_t + \\ + C_t^{\nabla\cdot\boldsymbol{J}}\rho_t\nabla\cdot\boldsymbol{J}_t + C_t^{\boldsymbol{J}^2}\boldsymbol{J}^2 \Big\}, \tag{3}$$

where the index t runs over 0 for isoscalar densities and 1 for isovector densities, respectively $\rho_0 = \rho_n + \rho_p$, $\rho_1 = \rho_n - \rho_p$ and similarly for all other quantities. p and n refer to protons and neutrons, respectively. The $C_t^{\rho}[\rho_0]$ coupling is further split into two contributions

$$C_t^{\rho} = C_{t0}^{\rho} + C_{tD}^{\rho}\rho_0^{\sigma}. \tag{4}$$

The different densities are defined starting from the density matrix in coordinate space

$$\rho_q(\boldsymbol{r}\sigma, \boldsymbol{r}'\sigma') = \left\langle a_q^{\dagger}(\boldsymbol{r}\sigma)a_q(\boldsymbol{r}'\sigma')\right\rangle, \tag{5}$$

where the index $\sigma = \pm 1$ runs over the up ($\sigma = 1$) and down ($\sigma = -1$) components of the spin. The spin trace of the density matrix reads

$$\rho_q(\boldsymbol{r}, \boldsymbol{r}') = \sum_{\sigma=\pm1} \rho_q(\boldsymbol{r}\sigma, \boldsymbol{r}'\sigma'), \tag{6}$$

and the spin density reads

$$\boldsymbol{s}_q(\boldsymbol{r}, \boldsymbol{r}') = \sum_{\sigma\sigma'=\pm1} \rho_q(\boldsymbol{r}, \boldsymbol{r}') \left\langle\sigma'\right| \boldsymbol{\sigma} \left|\sigma'\right\rangle, \tag{7}$$

where $\boldsymbol{\sigma}$ is the vector of Pauli matrices. The densities in Eq. 3 are then defined as

$$\rho_q = \rho_q(\boldsymbol{r}, \boldsymbol{r}')|_{\boldsymbol{r}=\boldsymbol{r}'}, \tag{8}$$

$$\tau_q = \nabla' \cdot \nabla\rho(\boldsymbol{r}, \boldsymbol{r}')|_{\boldsymbol{r}=\boldsymbol{r}'}, \tag{9}$$

$$J_{q,\mu\nu} = \frac{1}{2i}(\nabla_\mu - \nabla'_\mu)s_{q,\nu}(\boldsymbol{r}, \boldsymbol{r}')|_{\boldsymbol{r}=\boldsymbol{r}'}, \tag{10}$$

$$J_{q,\kappa}(\boldsymbol{r}) = \sum_{\mu\nu} \varepsilon_{\kappa\mu\nu} J_{q,\mu\nu}, \tag{11}$$

where $\varepsilon_{\kappa\mu\nu}$ is the Levi-Civita symbol. The spin-orbit density current tensor $\bar{\bar{J}}$ contribution to the EDF is generally split into its irreducible components. The scalar and tensor (irreps) parts vanish in our description, since they are time-odd [4]. The only non-zero contribution is given by the time-odd vector component $J_{q,\kappa}$ defined in Eq. 11.

2. *Kinetic energy and center of mass correction*

The kinetic energy term reads

$$\mathcal{E}_{\rm Kin} = \frac{\hbar^2}{2m}, \tag{12}$$

* Also at Physics Department, XYZ University.
† Contact author: Second.Author@institution.edu

while the loss of the center of mass symmetry loss is approximately compensated by the one-body center of mass correction correction term, which reads

$$\mathcal{E}_{\rm cm} = -\frac{1}{2mA}\sum_i \left\langle p_i^2 \right\rangle . \tag{13}$$

This equates to a renormalization of the nucleon mass, which then takes the form

$$m \to m\frac{A}{A-1}. \tag{14}$$

3. Coulomb energy

The Coulomb energy is treated within the Local Density Approximation. The resulting energy density reads [5]

$$\begin{aligned}\mathcal{E}_{\rm Coul} &= \mathcal{E}_{\rm Coul,\ D} + \mathcal{E}_{\rm Coul,\ E} \\ &= \frac{e^2}{2}\left[\int \frac{\rho_p(\boldsymbol{r})\rho_p(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}d\boldsymbol{r}' - \frac{3}{2}\left(\frac{3}{\pi}\right)^{\frac{1}{3}}\rho_p^{4/3}(\boldsymbol{r})\right].\end{aligned} \tag{15}$$

4. Pairing energy

The pairing interaction is assumed to be different from the particle-hole one, as done extensively in the literature. Treating the particle-particle channel requires the definition of the skew-symmetric pairing tensor

$$\kappa_q(\boldsymbol{r}\sigma, \boldsymbol{r}'\sigma') = \left\langle a_q(\boldsymbol{r}'\sigma')a_q(\boldsymbol{r}\sigma)\right\rangle , \tag{16}$$

which can be casted on a pairing density matrix that reads

$$\tilde{\rho}_q(\boldsymbol{r}\sigma, \boldsymbol{r}'\sigma') = -2\sigma'\kappa_q(\boldsymbol{r}\sigma, \boldsymbol{r}'\sigma'). \tag{17}$$

The local pairing density finally reads

$$\tilde{\rho}_q(\boldsymbol{r}) = \sum_{\sigma=\pm 1}\tilde{\rho}_q(\boldsymbol{r}\sigma, \boldsymbol{r}\sigma). \tag{18}$$

In this work, we employ the zero-range interaction which results in the energy density [6]

$$\mathcal{E}_{\rm Pair} = \sum_{q=n,p}\frac{1}{4}V_q\tilde{\rho}_q(\boldsymbol{r})\tilde{\rho}_q^*(\boldsymbol{r})\left[1-\eta\frac{\rho_0(\boldsymbol{r})}{\rho_s}\right]. \tag{19}$$

Here, $\eta \in [0,1]$ is a parameter which controls the spatial strength of the pairing interaction, $\eta = 0$ corresponds to volume pairing, $\eta = 1$ to surface pairing, and mixed pairing otherwise. ρ_s is the nuclear saturation density, fixed at 0.16 fm^{-3}.

B. HFB equations

Minimizing the EDF constraining the single-particle wavefunctions orthonormality and the particle number leads to the Hartree–Fock–Bogoliubov equations

$$\begin{pmatrix} h-\lambda & \Delta \\ -\Delta^* & -(h-\lambda)\end{pmatrix}\begin{pmatrix} U_\mu \\ V_\mu\end{pmatrix} = E_\mu \begin{pmatrix} U_\mu \\ V_\mu\end{pmatrix}, \tag{20}$$

where E_μ are the quasiparticle energies and U_μ and V_μ are the respective components of the quasiparticle wavefunctions. The mean-field Hamiltonian h and the pairing field Δ are obtained as variations of the functional, respectively $h_{ij} = \delta\mathcal{E}/\delta\rho_{ij}$ and $\Delta_{ij} = \delta\mathcal{E}/\delta\kappa_{ij}^*$. λ is a Langrange multiplier used to fix the particle number.

1. Mean-field Hamiltonian

The variation of the energy functional with respect to the density matrix ρ_{ij} leads to the single-particle Hamiltonian, which projected on the coordinate basis reads

$$h = \nabla\cdot\frac{\hbar^2}{2m^*(\boldsymbol{r})}\nabla + U_q(\boldsymbol{r}) + B_q(\boldsymbol{r})\cdot(\nabla\times\hat{\sigma}), \tag{21}$$

where the different fields in the equation are computed from the variation of the energy functional with respect to the local densities

$$U_q = \frac{\delta\mathcal{E}}{\delta\rho},\ B_q = \frac{\delta\mathcal{E}}{\delta\boldsymbol{J}},\ m^* = \frac{\delta\mathcal{E}}{\delta\tau}. \tag{22}$$

The self-consistent solution of Eq. 20 in coordinate space can be obtained in different ways. In this work, we employ the two-basis method, a detailed discussion can be found in Refs. [7–9].

C. Two-basis method

In the two-basis method, instead of diagonalizing the full HFB matrix, the problem is split into finding two bases: (a) the HF basis which diagonalizes the mean-field Hamiltonian h and (b) the quasiparticle basis, expressed in the HF basis, which diagonalizes the HFB Hamiltonian. In practice, this means that at each iteration, we solve the eigenvalue problem

$$h\left|\phi_k\right\rangle = \varepsilon_k\left|\phi_k\right\rangle , \tag{23}$$

which yields the HF basis $|\phi_k\rangle$. We can then compute the local pairing density (Eq. 18) as

$$\tilde{\rho}_q(\boldsymbol{r}) = \sum_{\sigma=\pm 1}\kappa_{ij}f_if_j\phi_i(\boldsymbol{r}\sigma)\phi_j(\boldsymbol{r}-\sigma), \tag{24}$$

The pairing gaps in this basis are then computed as

$$\Delta_{ij} = \sum_{\sigma=\pm 1}\int d\boldsymbol{r}\frac{V_q}{2}\left[1-\eta\frac{\rho_0(\boldsymbol{r})}{\rho_s}\right]\phi_i(\boldsymbol{r}\sigma)\phi_j(\boldsymbol{r}-\sigma)f_if_j. \tag{25}$$

The f_i, f_j factors are used to regularize the pairing interaction, as $\mathcal{E}_{\text{Pair}}$ is local in the pairing density it implies constant coupling in momentum space. To avoid sudden changes in the states considered, the factors used are the ones proposed in Ref. [10], which read

$$f_i = \left[\frac{1}{1+e^{(\varepsilon_i-\lambda_q-\Delta\varepsilon_q)/\mu_q}} \cdot \frac{1}{1+e^{(-\varepsilon_i+\lambda_q-\Delta\varepsilon_q)/\mu_q}}\right]^{1/2},$$

where $\Delta\varepsilon_q$ is the pairing window and μ_q is a factor which controls the sharpness of the window. After diagonalizing the HFB Hamiltonian in this basis, the density matrix and the pairing tensor are computed as

$$\rho = V^* V^T, \quad \kappa = U^* V^T. \tag{26}$$

The next mean field Hamiltonian is computed from the densities calculated on the canonical basis $|\psi_i\rangle$, i.e. the set of orbitals that diagonalizes the density matrix ρ

$$(S^{-1}\rho S)_{ij} = \delta_{ij} v_i^2. \tag{27}$$

The canonical basis, in the HF basis thus reads

$$|\psi_i\rangle = \sum_k S_{ki} |\phi_k\rangle . \tag{28}$$

D. The Lipkin-Nogami method

The Lipking-Nogami method (LN) [11] is an approximate variation after projection of the particle number. In this framework the total energy of the system becomes

$$E[\rho,\kappa] \mapsto E[\rho,\kappa] - \lambda_2(\langle N^2\rangle - \langle N\rangle^2). \tag{29}$$

The calculation of λ_2 is carried out at each iteration using the prescription [12, 13]:

$$\lambda_2 = \frac{G_{\text{eff}}}{4} \frac{\operatorname{Tr}\{(1-\rho)\kappa\}\operatorname{Tr}\{\rho\kappa\} - 2\operatorname{Tr}\{(1-\rho)^2\rho^2\}}{[\operatorname{Tr}(1-\rho)]^2 - 2\operatorname{Tr}\rho^2(1-\rho)^2}, \tag{30}$$

where G_{eff} is set to

$$G_{\text{eff}} = -\frac{\bar{\Delta}^2}{E_{\text{pair}}}. \tag{31}$$

Here, $\bar{\Delta}$ is the average pairing gap, computed as

$$\bar{\Delta} = \frac{\operatorname{Tr}\{\Delta\rho\}}{\operatorname{Tr}\rho}. \tag{32}$$

This phenomenological choice avoids the costly evaluation of the fields, which depends on the specific pairing interaction employed. On top of an approximate particle projection (before variation), the Lipkin-Nogami method introduces a curvature in the energy surface which prevents pairing collapse. This is important in deformed calculations, as certain spatial configurations along iterations may result in the trivial solution $\Delta = 0$, which is not recoverable in an HFB calculation. The substitution in (29) modifies the HFB matrix (20). Different choices can be made, depending on which expression for the particle dispersion is used [9]. In this work, the choice

$$h \mapsto h - 2\lambda_2(1-2\rho) \tag{33}$$

is used.

II. NUMERICAL TREATMENT

A. Numerical derivatives

Derivatives at each iteration are computed using the lagrange mesh scheme [14]. This enables reaching very high accuracies even for very modest meshes and guarantees that the method is variational. The single-particle Hamiltonian h is discretized with a five point scheme [15].

B. Implementation of the GCG method

The minimization of the energy functional on a cartesian mesh by repeated diagonalization of the single-particle Hamiltonian h is unfeasible. Even very modest discretizations lead to large-scale matrices which cannot be diagonlized exactly in reasonable CPU times. There exists a wide range of methods that can be employed for this kind of problem. One promising technique is the use of the preconditioned gradient descent [16, 17]. In this work we implement an algorithm based on the Generalized Conjugate Gradient (GCG) method [18, 19].

1. *Algorithm*

The method starts from a set of N_b guess single-particle wavefunctions $\{\phi_k^{(0)}\}$ and iteratively updates them until some convergence criteria is satisfied.

At iteration (i) of the procedure, a block W is constructed by an application of the inverse single-particle hamiltonian $h^{(i)}$, approximately solving the linear system

$$(h^{(i)} - \varepsilon_0^{(i)})W_k^{(i)} = \varepsilon_k^{(i)}\phi_k^{(i)}, \tag{34}$$

where $\varepsilon_0^{(i)}$ is the lowest single-particle energy at iteration (i). A conjugate search direction block is built using the previous iteration's states

$$P_k^{(i)} = \phi_k^{(i)} - \langle\phi_k^{(i-1)}|\phi_k^{(i)}\rangle\phi_k^{(i-1)}. \tag{35}$$

The new states for the iteration $(i+1)$ are finally obtained by diagonalizing $h^{(i)}$ in the vector space

$$\mathcal{V}^{(i)} = \operatorname{span}\{W^{(i)}, P^{(i)}, F^{(i)}\} \tag{36}$$

where $F^{(i)}$ is the block

$$F_k^{(i)} = \phi_k^{(i)}. \tag{37}$$

The diagonalization of $h^{(i)}$ in $\mathcal{V}^{(i)}$ is equivalent to solving the problem

$$V^{(i)\dagger} h^{(i)} V^{(i)} C = \Lambda C \tag{38}$$

which is of dimension $3N_b \times 3N_b$. $V^{(i)}$ is a matrix constructed by building the block $[W^{(i)}, P^{(i)}, F^{(i)}]$ and orthonormalizing it. Λ is a diagonal matrix containing the eigenvalues of the small eigenvalue problem and C is a matrix containing the eigenvectors. The new eigenvectors $\{\phi_k^{(i+1)}\}$ are the ones corresponding to the lowest N_b eigenvalues and they are obtained by

$$\phi_k^{(i+1)} = \sum_{j=1}^{N_b} \phi_j^{(i)} C_{kj}. \tag{39}$$

2. *Comparison with other algorithms*

As shown in Fig. 1 GCG performs better when compared with the very similar counterpart LOBPCG [20]. It is not only faster, but more robust and stable as well, while maintaining an identical computational effort. Moreover, a filtering operator is not necessary when variational collapse is a problem, e.g. in the covariant case [17] or the self-consistent solution of the HFB matrix by minimization [21]. Additionally, no search for an optimal preconditioner is needed, as the single-particle matrix is used in the inverse step.

Another interesting comparison can be done with the Imaginary Time Evolution (ITE) method, used for example by the Skyrme code EV8 [22]. While the computational overhead of ITE is very light, requiring only a matrix application and an orthonormalization of the eigenvectors, the execution time is dominated by the recalculation of the derivatives, blowing up the execution time: 159 s against the 21 s required by GCG.

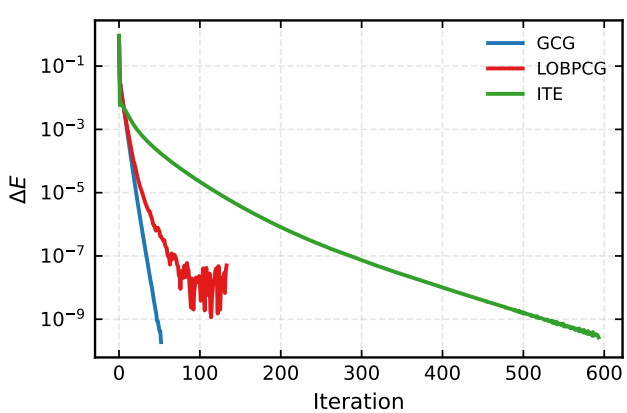


FIG. 1. Convergence of the total energy with three different algorithms. ^{40}Ca using SLy5. $\Delta E^{(i)} = |E^{(i)} - E^{(i-1)}|/|E^{(i)}|$.

C. Spatial constraints

Spatial constraints are implemented in this work by using the Augmented Lagrangian Method [23], which minimizes the Routhian

$$E' = E + \lambda^{(i)}(\langle Q \rangle - q) + \frac{c}{2}(\langle Q \rangle - q)^2, \tag{40}$$

where Q is a multipole operator, q a target value for said operator, and $\lambda^{(i+1)}$ is a Lagrange multiplier updated at each mean field iteration using the damped scheme

$$\lambda^{(i+1)} = \lambda^{(i)} + \mu c(\langle Q \rangle - q), \tag{41}$$

where $\mu \approx 0.05$ is a constant used to damp oscillations of the constraint field [22].

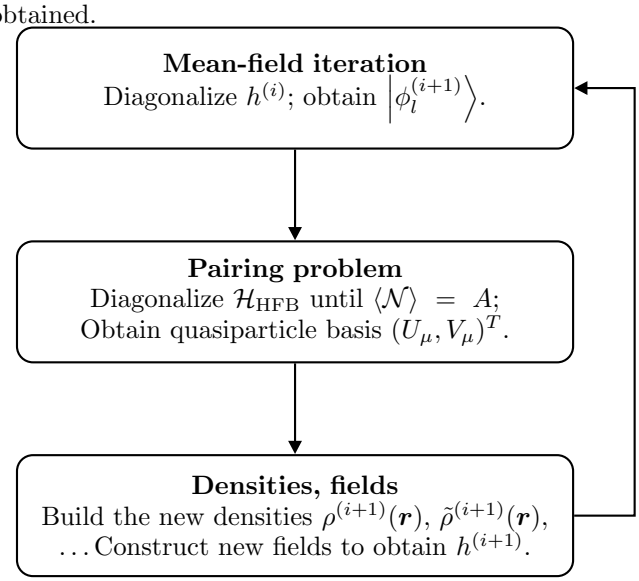
D. Particle number constraint

The lagrange multiplier λ in (20) is used to fix the particle number average $\langle N \rangle$. It is determined during the diagonalization of the HFB matrix using a bisection algorithm. This can be done by first determining a bracketing interval $[\lambda_{\min}, \lambda_{\max}]$ in which the correct value of λ shall lie. The bisection algorithm then proceeds as follows: (a) Diagonalize the HFB matrix

$$\begin{pmatrix} h - \lambda^{(j)} & \Delta \\ -\Delta^* & -(h - \lambda^{(j)}) \end{pmatrix} \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix}^{(j)} = E_\mu \begin{pmatrix} U_\mu \\ V_\mu \end{pmatrix}^{(j)}; \tag{42}$$

(b) determine the particle number $\langle N \rangle^{(j)}$; (c) if $\langle N \rangle^{(j)} < A$, set $\lambda_{\min} = \lambda^{(j)}$; otherwise set $\lambda_{\max} = \lambda^{(j)}$; (d) set $\lambda^{(j+1)} = (\lambda_{\min} + \lambda_{\max})/2$. Repeat until convergence.

The superscript (j) is used to avoid confusion between the outer mean-field iterations (i) and the bisection iterations (j). During the bisection algorithm, only the $\lambda^{(j)}$ value is updated, for which a new quasiparticle basis is obtained.



Appendix A: Local densities and fields

We report here the formulae for the local densities in coordinate space

$$\rho_q(\boldsymbol{r}) = \sum_{k,\sigma} v_k^2 |\psi(\boldsymbol{r}\sigma)|^2, \tag{A1}$$

$$\tau_q(\boldsymbol{r}) = \sum_{k,\sigma} v_k^2 |\nabla\psi(\boldsymbol{r}\sigma)|^2, \tag{A2}$$

$$J_{q,\mu\nu}(\boldsymbol{r}) = \sum_{k,\sigma\sigma'} v_k^2 \mathrm{Im}\{\psi(\boldsymbol{r}\sigma)\partial_\mu \sigma_{\nu,\sigma\sigma'}\psi(\boldsymbol{r}\sigma')\}, \tag{A3}$$

$$J_{q,\kappa}(\boldsymbol{r}) = \sum_{\mu\nu} \varepsilon_{\kappa\mu\nu} J_{q,\mu\nu}. \tag{A4}$$

Here, the index k runs over the canonical basis wavefunctions. v_k^2 denotes the occupation factor of the wavefunction k computed by diagonalizing the density matrix.

Appendix B: Mean-field

The central field reads

$$\begin{aligned} U_q = &(2C_0^\rho - 2C_1^\rho)\,\rho + 4C_1^\rho\,\rho_q + \\ &(C_0^\tau - C_1^\tau)\,\tau + 2C_1^\tau\,\tau_q + \\ &(2C_0^{\Delta\rho} - 2C_1^{\Delta\rho})\,\nabla^2\rho + 4C_1^{\Delta\rho}\,\nabla^2\rho_q + \\ &(C_0^{\nabla J} - C_1^{\nabla J})\,\nabla\cdot\mathbf{J} + 2C_1^{\nabla J}\,\nabla\cdot\mathbf{J}_q + \\ &(\sigma+2)C_{0D}^\rho\rho^{\sigma+1} + \\ &C_{1D}^\rho\rho^{\sigma-1}(\sigma\rho_p^2 + \sigma\rho_n^2 + 2\sigma\rho_p\rho_n + 2\rho_p^2 - 2\rho_n^2), \end{aligned}$$

the spin-orbit field reads

$$\begin{aligned} \boldsymbol{B}_q = &2(C_0^{\boldsymbol{J}^2} - C_1^{\boldsymbol{J}^2})\boldsymbol{J} + 4C_1^{\boldsymbol{J}^2}\boldsymbol{J}_q - \\ &(C_0^{\nabla\cdot\boldsymbol{J}} - C_1^{\nabla\cdot\boldsymbol{J}})\nabla\rho - 2C_1^{\nabla\cdot\boldsymbol{J}}\nabla\rho_q, \end{aligned}$$

and the effective mass field reads

$$\begin{aligned} \frac{\hbar^2}{2m^*} = &\frac{\hbar^2}{2m} + \\ &(C_0^\tau - C_1^\tau)\rho + 2C_1^\tau\rho_q + \\ &(C_0^\tau - C_1^\tau)\nabla\rho + 2C_1^\tau\nabla\rho_q. \end{aligned} \tag{B1}$$

1. Coulomb field

The direct part of the Coulomb field, i.e.

$$U_{c,D} = \frac{e^2}{2}\int d\boldsymbol{r}'\,\frac{\rho_p(\boldsymbol{r}')}{\|\boldsymbol{r}' - \boldsymbol{r}\|}, \tag{B2}$$

is computed by solving the Poisson equation

$$\nabla^2 U_{c,D} = -4\pi e^2 \rho_p \tag{B3}$$

to avoid the costly evaluation of the integral. The energy density is then set to

$$\mathcal{E}_{\text{Coul, D}} = U_{c,D}\rho_p. \tag{B4}$$

The exchange field reads

$$U_{c,E} = \frac{\delta\mathcal{E}_{\text{Coul, E}}}{\delta\rho} = -e^2\left(\frac{3}{\pi}\right)^{1/3}\rho_p^{1/3}. \tag{B5}$$

2. Constraint field

If constraints are present, the constraint field reads

$$U_{\text{Constr}} = \lambda^{(i)}Q + c(\langle Q\rangle - q)Q. \tag{B6}$$

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